

Self-improving System Integration – On a Definition and Characteristics of the Challenge

Kirstie Bellman*, Christian Gruhl[§], Chris Landauer*, and Sven Tomforde[§]

*Topcy House Consulting, US, Email: bellmanhome@yahoo.com, topcycal@gmail.com

[§]Intelligent Embedded Systems, University of Kassel, Germany, Email: {cgruhl|stomforde}@uni-kassel.de

Abstract—The self-improving system integration (SISSY) initiative—sometimes also called self-integration—started over six years ago. Recently, we have seen contributions approaching parts of the scope from different directions. In order to provide a better understanding how these attempts support the development of novel SISSY technology, this paper presents a definition of the term. In addition, it discusses the fundamental challenges stemming from the increasingly interconnected character of large-scale distributed and adaptive systems typically referred to as “Interwoven Systems”, with these systems describing the basic motivation for SISSY research.

I. INTRODUCTION

The term “integration” describes the combination of parts towards a final product or system. However, integration is more than just putting things together. Especially in the context of large-scale systems consisting of a variety of heterogeneous, autonomous subsystems, the integration task turns out to become a nightmare for system engineers [1].

Integration in such complex systems has always been technically difficult; traditionally requiring extensive design-time analysis and testing to make sure that the desirable properties of a configuration of components results in the appropriate overall system performance and to ensure that undesirable side-effects of the integration do not occur or, at least, are mitigated. This challenge is even more difficult when we shift to the need for more autonomous reconfiguration and runtime adaptations of systems in highly dynamic and variable environments. Large, distributed networks of both hardware and software capabilities are expected to respond at runtime with increasingly adaptive and autonomous behaviours that act in a coordinated fashion.

In particular, initiatives such as Proactive Computing (PAC) [2], Autonomic Computing (AC) [3], or Organic Computing (OC) [4] emerged in response to the ever increasing complexity of software and hardware systems. These initiatives propose to tackle the observed growth in complexity by means of autonomous behaviour, i.e., shifting decision freedom from the engineer to the systems themselves. However, this also resulted in reduced predictability of system behaviour and consequently to even more challenges for integration tasks. In turn, the self-improving system-integration initiative claims that the most promising solution for mastering the reduced predictability and the corresponding challenges again lies in allowing for autonomous behaviour by transferring even more integration decisions to the systems and into the runtime.

In this paper, we discuss first what exactly self-integration in complex systems means, and try to come up with a definition of the term “self-improving system integration” (Section II). Afterwards, we discuss the most urgent challenges for self-integration based on the notion of “Interwoven Systems” (Section III).

II. DEFINITION

We characterise “self-improving system integration” (SISSY) as an attempt for *intelligent* integration of complex systems that have both adaptive and non-adaptive components or subsystems. In the following paragraphs, we approach a more precise definition of the term by initially revisiting the notion of “integration” in traditional engineering, followed by a discussion of the “self” and the “improving” parts.

In engineering, the notion of integration describes a process in which several component (sub-)systems are brought together and interconnected into a unified system. This aims to achieve a correctly working unit, where the subsystems work together to provide desired functions, with acceptable performance and dependability properties. In classic engineering, the integration of (sub-)systems has been done at design-time, with rigid specifications and testing of (sub-)system interfaces and performance. However, with increasingly networked and open systems (e.g., internet of things [5], smart homes and grids [6], or smart traffic [7], [8]), we now face the challenge of integrating systems dynamically, as rapidly as possible. Because of the dynamic contexts—where goals, resources, and knowledge required for integration change rapidly—increasing efforts have been directed towards new processes and computations that allow intelligent systems to do most of the integration themselves.

Hence, we define “self-integration” as an ongoing autonomous process for linking a potentially large set of heterogeneous computing systems, devices, and software applications; so as to meet system goals. The linking itself is done physically or functionally. We consider this process to be continual, i.e., it is never finished, as the way in which the integrated elements—software and hardware—act together must adapt to external changes, goal evolution, and autonomous decisions of these elements [9]. The goals to be achieved are typically not only covering a single objective—in turn they may comprise several (possibly conflicting) aspects. There may even be a deviation between goals of the overall

system and the individual component subsystems, see [10] or [11].

Within this process, two major tasks are performed: (i) connections between subsystems are established, tested, and assessed; and, (ii) the particular subsystems in combination with their behavioural strategies are configured depending on the operational and functional area. This has a strong overlap with the concept of "self-organisation" [12]: (i) connecting subsystems or elements refers to building the system's structure; and, (ii) configuration is mainly concerned with defining the desired behaviour. It also relies on concepts from self-aware computing systems [13] that use internal knowledge and learning processes to reason about their resources and state. Multi-agent systems also provide relevant concepts in terms of organisational paradigms [14], where agents can cooperate or compete, forming hierarchies, coalitions, teams, federations, societies and so on [15].

As mentioned before, self-integration is a continuing process rather than a single event task, and it is a relatively open process. Since not all conditions can be anticipated, the process needs a "creative component" that is responsible for inventing solutions where none are available and for optimising the selection of solutions over time. We refer to this concept as "self-improvement". It implies that self-integrating systems do not settle for the status quo, but rather rely on continuous learning to optimise their behaviours with respect to their goals; even though they may or may not be able to reach optimal states or behaviours. Such a self-improvement process requires a utility function that guides the optimisation process towards desired states. However, this utility is not necessarily a single objective—in turn, it may be composed of several even conflicting aspects, as highlighted in [10], requiring negotiation processes to make the appropriate trade-offs for any given situation.

III. CHARACTERISTICS OF THE Sissy CHALLENGE

Self-integration in combination with self-improvement mechanisms has been established as a research initiative in response to current trends in information and communication technology. Especially, characteristics such as openness and interconnectedness of large-scale collections of subsystems demand for novel answers. Driven by this ongoing trend, e.g. also expressed by the domains of internet-of-things [5] or cyber-physical systems [16], the term "Interwoven Systems" (IwS) [17], [18], [19] has been coined to describe a class of systems comprising these challenges.

According to [20], IwS are defined as follows: "IwS are systems where as many elements (subsystems) as possible are equipped with interweaving capabilities. In general, the overall system has the following properties: (1) *mutual influences* (i.e., there exist a number of interactions not defined at design-time that shall be made explicit at runtime), (2) *heterogeneity of system elements* (i.e. system elements may belong to different application domains or authorities), (3) *uncertainty* (i.e., the predictability of the entire system and its parts is almost *always* incomplete and inaccurate), and (4) soft or hard real-time

aspects (i.e., elements must operate complying to deadlines). The interweaving capabilities of an element are based on techniques for (i) online dependency detection and modelling to reveal the various kinds of interactions, (ii) online goal adaptation (e.g., goal negotiation or re-weighting of subgoals), (iii) continuous re-design (e.g., parameters or internal structure of an element), and (iv) long-term self-improvement". [21]

This definition builds upon Maier's concept of "System-of-Systems" [22], but adds the essential features of self-directed integration. However, it already outlines basic challenges of IwS that are especially relevant to be covered by Sissy mechanisms. We briefly outline the most important ones in the following paragraphs.

Mutual influences: A Sissy system acts within the frame of an overall system-of-systems structure [17] and a shared environment. Consequently, actions of one Sissy system may have impact on the other systems in the vicinity, which may affect their utility function. In order to avoid negative effects in terms of disturbing the learning behaviour and/or decreasing the utility, systems need the ability to discover such mutual influences at runtime. A corresponding methodology has been proposed by Rudolph et al. in [23].

Heterogeneity: As we cannot expect that all elements are accessible to equip them with interweaving capabilities, the goal is to reduce failures in operation or even the chance of catastrophic outages as far as possible and to reduce costs in terms of system development and operation. We can imagine helper processes that shepherd these less capable elements through the integration processes [24].

Metrics: Assessing integration quality requires meaningful metrics, e.g., based on the categorisation and levels as introduced by [11]. Within these categories, several fundamental qualities such as reliability, performance, availability, interoperability, scalability and consistency are distinguished. However, there are also more integration-specific, aggregate measures such as system adaptability or integration efficiency, see [10]. The challenge here is to define appropriate metrics for quantifying the integration status, which may also be situation dependent.

Flexibility: Closely related to the underlying metrics for success estimation and quantifying the integration status is flexibility: The possibility to adapt these goals and the corresponding utility functions at runtime. This can be either triggered by the user/owner of the system or by the system itself as response to available resources or the internal/external status, for instance. Technically, this is especially challenging for the self-improving part of Sissy systems, since learning builds models based on the utility and changing the utility requires updating existing knowledge, see e.g. [25].

Autonomous Learning: The self-improvement capability requires autonomous learning mechanisms. In contrast to traditional approaches (e.g., realised as basic reinforcement methods [26]), more sophisticated techniques are needed that autonomously assess their own knowledge, identify and close existing knowledge gaps, or detect changes in the underlying fitness landscape [27], [28].

Collaborative Interactive Learning: Besides autonomous learning, the systems may opportunistically use available knowledge sources (e.g., crowd sourcing mechanisms, the Internet, data bases) for learning purposes, and they may collaborate with other trustworthy component subsystems to share and assess knowledge [29].

Computational Trust: A SISSY system is considered to act within an *open* system-of-systems structure [17] and a shared environment. This means that component systems are free to join and leave. As a consequence, there are always new systems with unknown behaviour and there may also be malevolent systems. In order to be able to isolate negative component systems (e.g., attackers or malfunctioning systems), computational trust is a possible approach. This requires metrics and mechanisms as proposed in [30] and may establish closed communities with mutual trust relationships [31].

IV. CONCLUSION

In response to six years of research in the self-improving systems integration (SISSY) community, this paper proposed a definition of the term SISSY. We briefly summarised what integration and self-improvement mean in large-scale adaptive systems. Afterwards, we shifted the focus towards the specific characteristics of the SISSY challenges. Since an individual SISSY system is acting within an overall complex structure, this overall structure is highly important for acting successfully. We revisited the term “Interwoven System” as description of a class of interconnected system-of-systems organisations with specific properties. Based on these properties, we discussed the most prominent research challenges for actually implementing SISSY systems with the desired behaviour.

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